Analysis_Jan2025

2025-12-09

```r
library(data.table)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:data.table':
##
##     between, first, last

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)
```

```r
# Change this to the correct path (I'll fix this so people don't have to change this)
project_path <- "/Users/nencheva/Google Drive/My Drive/MasterFolder/Projects/Active/MB4/MB4 Affect Spin

base_path <- paste0(project_path,"/Data/")

affect_ratings <- data.table(read.csv(paste0(base_path,'affect_ratings_merged_with_marker_id.csv')))

main_data <- data.table(read.csv(paste0(base_path,'clean_data.csv'))) %>% select(lab_id,subj_id,age_days
       push_up_identity, push_up_order, push_up_side, meet_eligibility)


merged_data <- data.table(merge(affect_ratings,main_data, by = c('lab_id','subj_id')))
```

```r
trial_level_affect <- merged_data[
  ,
  {
    # total duration of coded window for this trial
    total_dur <- 10#sum(duration_sec, na.rm = TRUE)

    # duration (in sec) for positive, negative and any non-neutral
    non_neutral_dur <- sum(duration_sec[!is.na(affect_rating_segment) &
                                          affect_rating_segment != 4],
```

```r
                        na.rm = TRUE)
pos_dur <- sum(duration_sec[!is.na(affect_rating_segment) &
                              affect_rating_segment > 4],
               na.rm = TRUE)
neg_dur <- sum(duration_sec[!is.na(affect_rating_segment) &
                              affect_rating_segment < 4],
               na.rm = TRUE)

# centered affect (neutral = 0)
affect_centered <- affect_rating_segment - 4

# duration-weighted affect rating average INCLUDING neutral
affect_avg <- if (total_dur > 0) {
  sum(affect_centered * duration_sec, na.rm = TRUE) / total_dur
} else {
  NA_real_
}

# duration-weighted average EXCLUDING neutral (0s -> NA)
affect_avg_nonneutral <- {
  affect_centered_nz <- fifelse(affect_centered == 0, NA_real_, affect_centered)

  nonneutral_dur <- sum(duration_sec[!is.na(affect_centered_nz)], na.rm = TRUE)

  if (nonneutral_dur > 0) {
    sum(affect_centered_nz * duration_sec, na.rm = TRUE) / nonneutral_dur
  } else {
    NA_real_
  }
}


# get first reaction
first_idx_any <- {
  ok <- which(!is.na(segment_start_sec) & !is.na(affect_centered) & !is.na(duration_sec))
  if (length(ok) == 0) NA_integer_ else ok[which.min(segment_start_sec[ok])]
}

# find first affect reaction
first_affect <- if (!is.na(first_idx_any)) affect_centered[first_idx_any] else NA_real_




# Peak positive / negative ratings within trial (exploratory measure)
peak_pos <- {
  v <- affect_centered[!is.na(affect_centered)]
  if (length(v) == 0) NA_real_ else max(v)
}

peak_neg <- {
  v <- affect_centered[!is.na(affect_centered)]
  if (length(v) == 0) NA_real_ else min(v)
```

```r
}

# Strongest affect regardless of sign
peak_abs <- {
  v <- affect_centered[!is.na(affect_centered)]
  if (length(v) == 0) NA_real_ else max(abs(v))
}

peak_signed <- {
  v <- affect_centered[!is.na(affect_centered)]
  if (length(v) == 0) {
    NA_real_
  } else {
    v[which.max(abs(v))]   # keeps the sign of the strongest deviation
  }
}




# take first values per trial for video-level categorical ratings
happiness_val <- first(happiness_full_video)
surprise_val  <- first(surprise_full_video)

.(
  # keep other trial-level columns (take first value within trial)
  folder                  = first(folder),
  file_name               = first(file_name),
  condition               = first(condition),
  cb_order                = first(cb_order),
  helper_hinderer_choice  = first(helper_hinderer_choice),
  push_up_identity        = first(push_up_identity),
  push_up_order           = first(push_up_order),
  push_up_side            = first(push_up_side),
  meet_eligibility        = first(meet_eligibility),

  #affectPresent_duration: this is the cumulative duration of non-neutral expressions in seconds ov
  affectPresent_duration = if (total_dur > 0) non_neutral_dur / total_dur else NA_real_,

  #affectPositive_duration: this is the cumulative duration of positive expressions in seconds over
  affectPositive_duration = if (total_dur > 0) pos_dur / total_dur else NA_real_,

  #affectNegative_duration: this is the cumulative duration of negative expressions in seconds over
  affectNegative_duration = if (total_dur > 0) neg_dur / total_dur else NA_real_,

  #affect_average: this is the weighted average of the Likert ratings over the analysis window, thi
  affect_average = affect_avg,
  affect_average_nonneutral = affect_avg_nonneutral,
  # affect rating of first reaction for the trial (exploratory measure)
  first_affect = first_affect,
  # peak affect rating for the trial (exploratory measure)
  peak_affect_rating_abs = peak_abs,
  peak_affect_rating_signed = peak_signed,
```

```r
      #affectPositive_categoricalChoice: this is the categorical choice affect rating, coded as 1 for p
      affectPositive_categoricalChoice = if (is.na(happiness_val)) NA_integer_ else if (happiness_val ==
      #affectNegative_categoricalChoice: this is the categorical choice affect rating, coded as 1 for n
      affectNegative_categoricalChoice = if (is.na(happiness_val)) NA_integer_ else if (happiness_val ==
      #surprise_categoricalChoice: this is the categorical choice surprise rating, coded as 1 for surpr
      surprise_categoricalChoice = if (is.na(surprise_val)) NA_integer_ else if (surprise_val == 1) 1L
    )
  },
  by = .(lab_id, subj_id, trial_num)
]



# tag push up and push down trials
trial_level_affect[, trial_num_int := as.integer(trial_num)]
trial_level_affect[
  ,
  is_push_up :=
    (push_up_order == "first"  & trial_num_int %% 2L == 1L) |
    (push_up_order == "second" & trial_num_int %% 2L == 0L)
]

trial_level_affect[
  ,
  push_direction := ifelse(is_push_up, "push_up", "push_down")
]
trial_level_affect[, trial_num_int := NULL]


# Rename variables to align them with preregistered models
trial_level_affect[, social := factor(condition)]
trial_level_affect[
  ,
  helping := factor(
    ifelse(is.na(is_push_up), NA_character_,
           ifelse(is_push_up, "help", "hinder")),
    levels = c("hinder", "help")
  )
]
trial_level_affect[
  ,
  chose_helper := fifelse(
    helper_hinderer_choice == "helper",   1L,
    fifelse(helper_hinderer_choice == "hinderer", 0L, NA_integer_)
  )
]
```

```r
library(lmerTest)
```

```
## Loading required package: lme4

## Loading required package: Matrix
```

```
##
## Attaching package: 'lmerTest'

## The following object is masked from 'package:lme4':
##
##     lmer

## The following object is masked from 'package:stats':
##
##     step
```

```r
library(papaja)
```

```
## Loading required package: tinylabels
```

```r
# affectPresent_duration ~ social + (1 | participant_id) + (1 | lab_id)
m_affectPresent_social <- lmer(
  affectPresent_duration ~ social + (1 | subj_id) + (1 | lab_id),
  data = trial_level_affect
)





m_affectPresent_social_first2trials <- lmer(
  affectPresent_duration ~ social + (1 | subj_id) + (1 | lab_id),
  data = trial_level_affect[trial_num <= 2 ,]
)


p_affectPresent_social <- ggplot(
  trial_level_affect,
  aes(x = social, y = affectPresent_duration)
) +
  geom_jitter(width = 0.2, height = 0.1, alpha = 0.1,size =0.5) +
  stat_summary(
    fun.data = mean_cl_boot,
    geom = "pointrange",
    size = 0.2
  ) +
  theme_minimal() +
  labs(
    x = "Condition",
    y = "Affect present (duration proportion of trial)"
  )


# affect_average ~ social + (1 | participant_id) + (1 | lab_id)
m_affectAverage_social <- lmer(
  affect_average ~ social + (1 | subj_id) + (1 | lab_id),
  data = trial_level_affect
```

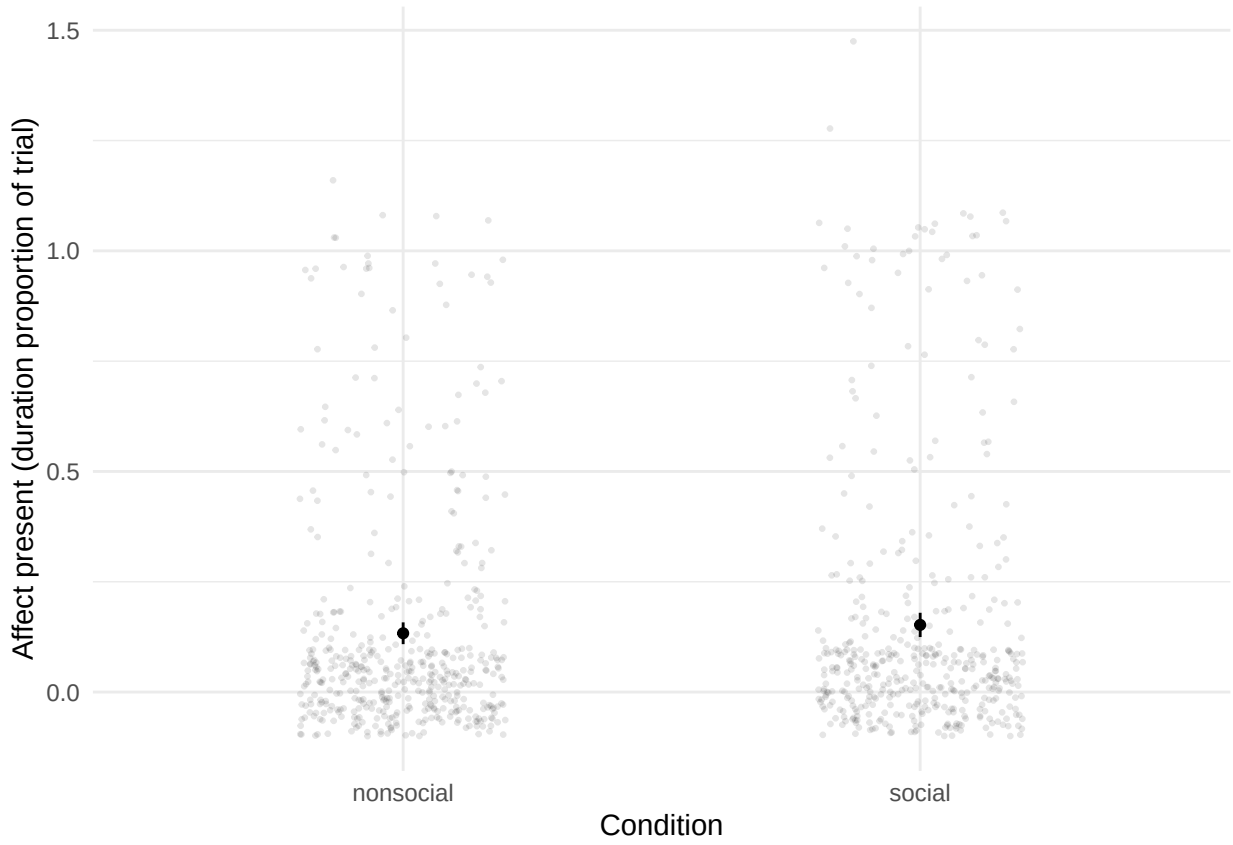
```r
)


m_affectAverage_social_first2trials <- lmer(
  affect_average ~ social + (1 | subj_id) + (1 | lab_id),
  data = trial_level_affect[trial_num <= 2 ,]
)



#excluding neutral affect
m_affectAverage_social_nonneutral <- lmer(
  affect_average_nonneutral ~ social + (1 | subj_id) + (1 | lab_id),
  data = trial_level_affect
)



p_affectAverage_social <- ggplot(
  trial_level_affect,
  aes(x = social, y = affect_average)
) +
  geom_jitter(width = 0.2, height = 0.1, alpha = 0.1,size =0.5) +
  stat_summary(
    fun.data = mean_cl_boot,
    geom = "pointrange",
    size = 0.2
  ) +
  theme_minimal() +
  labs(
    x = "Condition",
    y = "Average affect rating"
  )

print(paste("affectPresent_duration ~ social:", apa_print(m_affectPresent_social)$full_result$socialsoci
```

```
## [1] "affectPresent_duration ~ social: $\\hat{\\beta} = 0.02$, 95\\% CI $[-0.04, 0.08]$, $t(92.17) = (
```

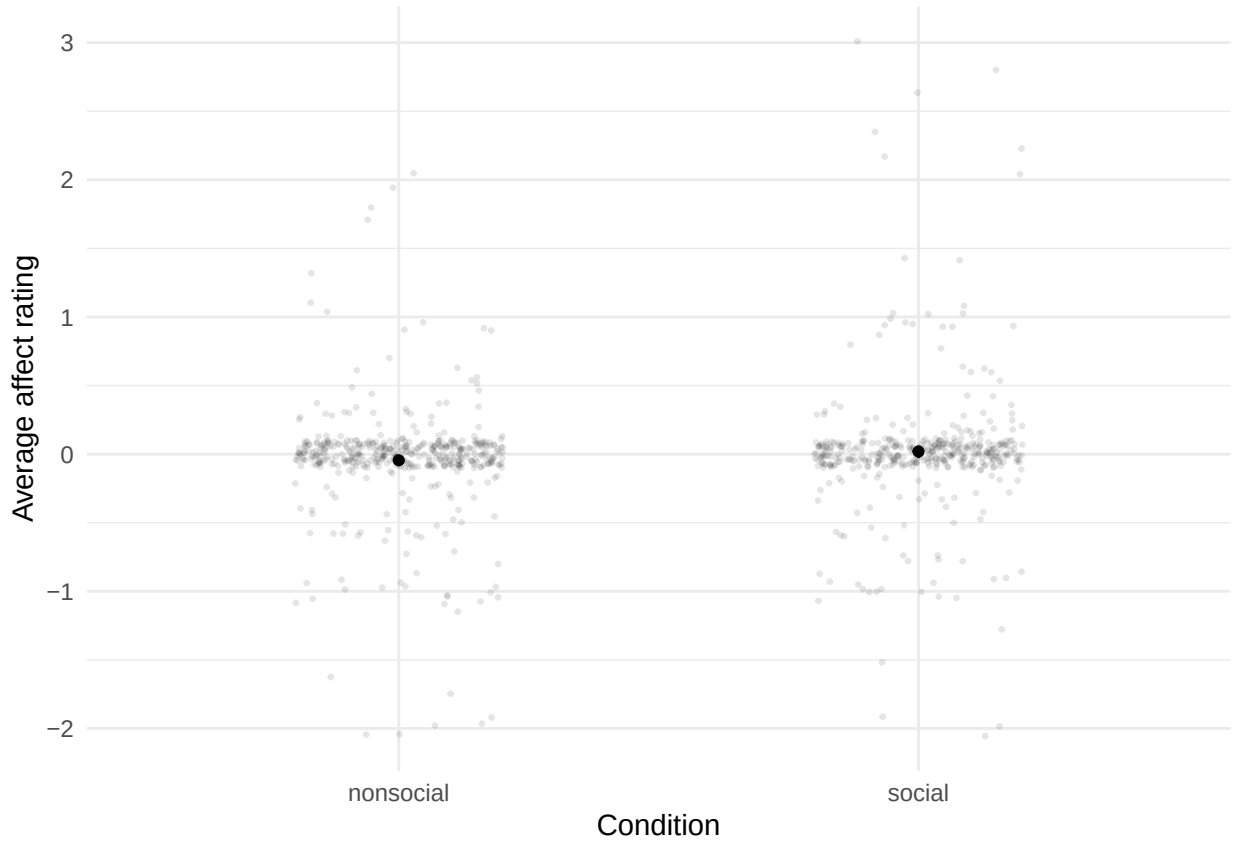
```r
print(paste("affect_average ~ social:", apa_print(m_affectAverage_social)$full_result$socialsocial))
```

```
## [1] "affect_average ~ social: $\\hat{\\beta} = 0.05$, 95\\% CI $[-0.04, 0.14]$, $t(95.53) = 1.14$, $p
```

```r
p_affectPresent_social
```




```
p_affectAverage_social
```




```
# affect_average ~ helping*social + (helping | participant_id) + (helping | lab_id)
m_affectAverage_helping_social <- lmer(
  affect_average ~ helping * social +
    (helping | subj_id) +
    (1 | lab_id),
  data = trial_level_affect
)
```

```
## boundary (singular) fit: see help('isSingular')
```

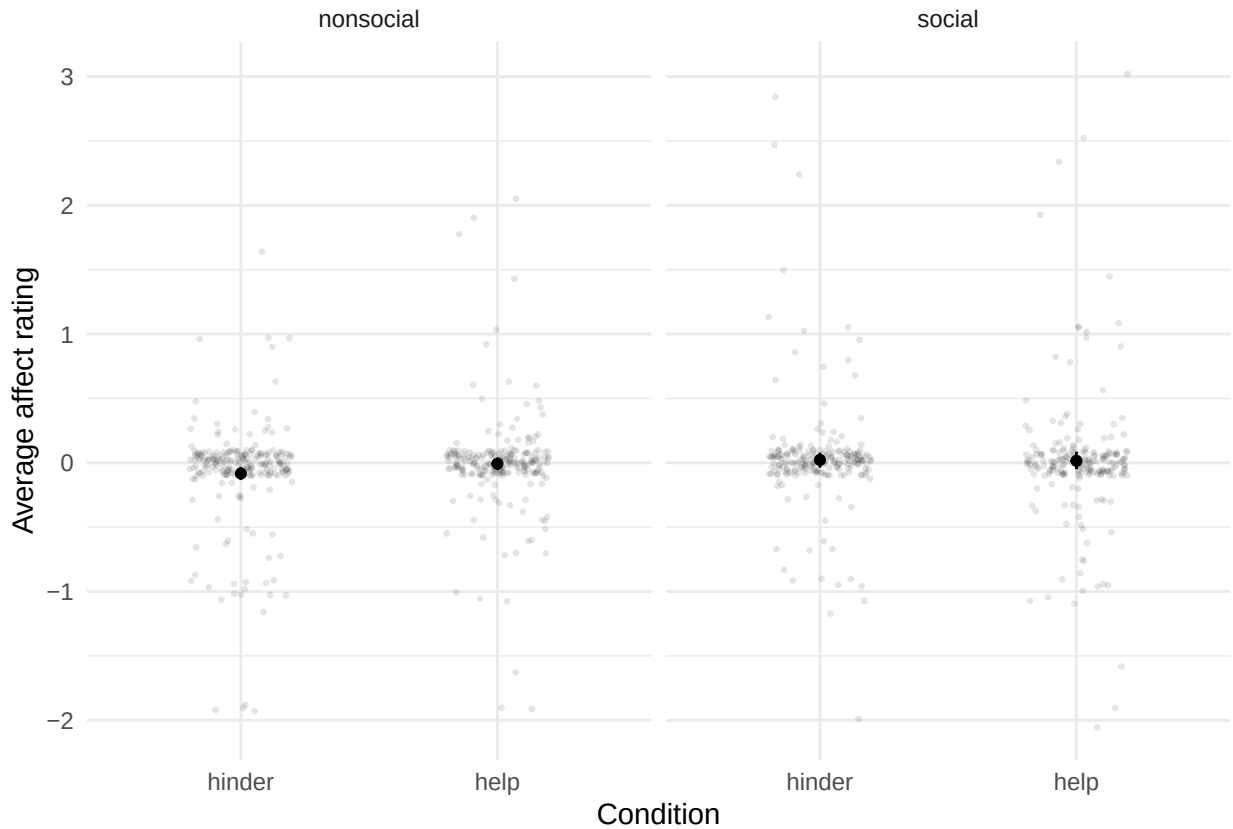
```
m_affectAverage_helping_social_first2 <- lmer(
  affect_average ~ helping * social +
    (1 | subj_id) +
    (1 | lab_id),
  data = trial_level_affect[trial_num <= 2,]
)


print("affect_average ~ helping*social:")
```

```
## [1] "affect_average ~ helping*social:"
```

```
apa_print(m_affectAverage_helping_social)$table
```

```
## A data.frame with 6 labelled columns:
##
##                                  term estimate      conf.int statistic     df
## 1                           Intercept    -0.08 [-0.17, 0.01]     -1.71   7.47
## 2                         Helpinghelp     0.07  [0.00, 0.15]      1.96 785.65
## 3                        Socialsocial     0.10 [-0.01, 0.20]      1.81 107.51
## 4 Helpinghelp $\\times$ Socialsocial    -0.09 [-0.19, 0.02]     -1.60 785.81
##   p.value
## 1    .128
## 2    .051
## 3    .072
## 4    .111
##
## term     : Term
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $t$
## df       : $\\mathit{df}$
## p.value  : $p$
```

```
p_affectAverage_helping <- ggplot(
  trial_level_affect,
  aes(x = helping, y = affect_average)
) +
  facet_grid(cols = vars(social))+
  geom_jitter(width = 0.2, height = 0.1, alpha = 0.1,size =0.5) +
  stat_summary(
    fun.data = mean_cl_boot,
    geom = "pointrange",
    size = 0.2
  ) +
  theme_minimal() +
  labs(
    x = "Condition",
    y = "Average affect rating"
  )

p_affectAverage_helping
```




```r
# affectPositive_duration ~ helping*social + (helping | participant_id) + (helping | lab_id)
m_affectPositive_helping_social <- lmer(
  affectPositive_duration ~ helping * social +
    (1 | subj_id),
  data = trial_level_affect
)

print("affectPositive_duration ~ helping*social:")
```

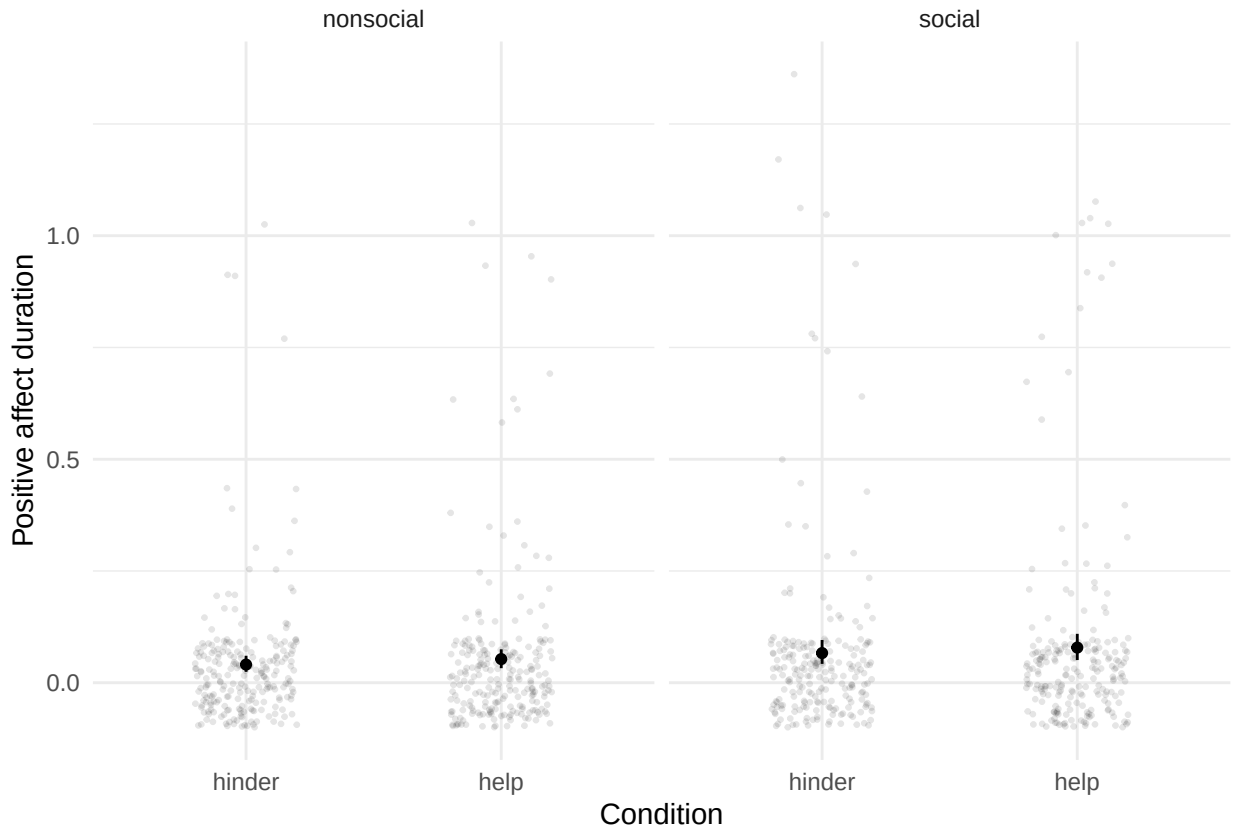
```
## [1] "affectPositive_duration ~ helping*social:"
```

```r
apa_print(m_affectPositive_helping_social)$table
```

```
## A data.frame with 6 labelled columns:
##
##                                 term estimate      conf.int statistic     df
## 1                          Intercept     0.04  [0.01, 0.07]      2.58 160.76
## 2                        Helpinghelp     0.01 [-0.02, 0.04]      0.73 814.36
## 3                       Socialsocial     0.02 [-0.02, 0.07]      0.96 165.92
## 4 Helpinghelp $\\times$ Socialsocial     0.00 [-0.05, 0.04]     -0.10 814.52
##   p.value
## 1    .011
## 2    .465
## 3    .337
```

```
## 4    .923
##
## term     : Term
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $t$
## df       : $\\mathit{df}$
## p.value  : $p$
```

```r
p_affectPositive_helping <- ggplot(
  trial_level_affect,
  aes(x = helping, y = affectPositive_duration)
) +
  facet_grid(cols = vars(social))+
  geom_jitter(width = 0.2, height = 0.1, alpha = 0.1,size =0.5) +
  stat_summary(
    fun.data = mean_cl_boot,
    geom = "pointrange",
    size = 0.2
  ) +
  theme_minimal() +
  labs(
    x = "Condition",
    y = "Positive affect duration"
  )


p_affectPositive_helping
```




```
# affectNegative_duration ~ helping*social + (helping | participant_id) + (helping | lab_id)
m_affectNegative_helping_social <- lmer(
  affectNegative_duration ~ helping * social +
    (1 | subj_id) +
    (1 | lab_id),
  data = trial_level_affect
)





print("affectNegative_duration ~ helping*social:")
```

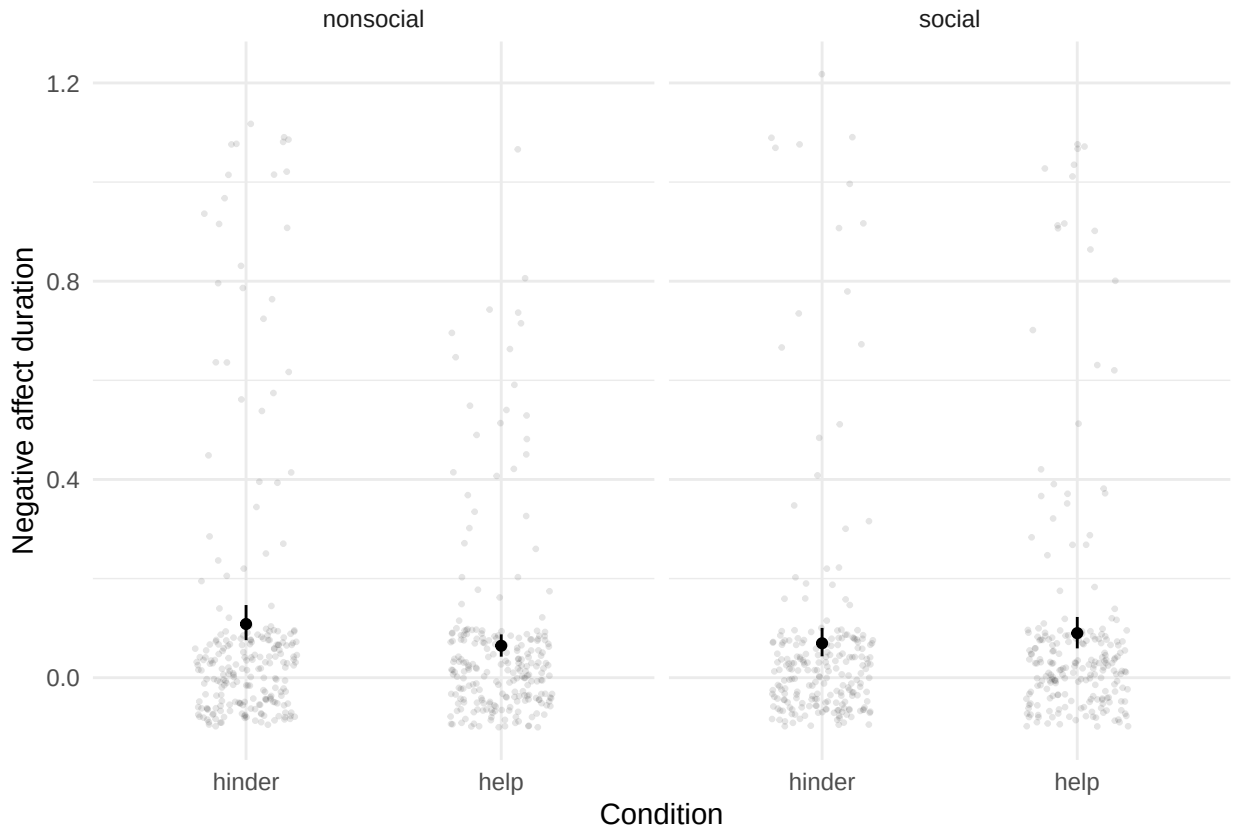
```
## [1] "affectNegative_duration ~ helping*social:"
```

```
apa_print(m_affectNegative_helping_social)$table
```

```
## A data.frame with 6 labelled columns:
##
##                                term estimate       conf.int statistic     df
## 1                          Intercept     0.11   [0.05, 0.17]      3.53   5.20
## 2                       Helpinghelp    -0.04 [-0.08, -0.01]     -2.26 811.43
```

```
## 3                        Socialsocial    -0.04  [-0.09, 0.02]     -1.38 166.04
## 4 Helpinghelp $\\times$ Socialsocial     0.07   [0.01, 0.12]      2.32 811.70
##   p.value
## 1    .016
## 2    .024
## 3    .169
## 4    .020
##
## term     : Term
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $t$
## df       : $\\mathit{df}$
## p.value  : $p$
```

```
p_affectNegative_helping <- ggplot(
  trial_level_affect,
  aes(x = helping, y = affectNegative_duration)
) +
  facet_grid(cols = vars(social))+
  geom_jitter(width = 0.2, height = 0.1, alpha = 0.1,size =0.5) +
  stat_summary(
    fun.data = mean_cl_boot,
    geom = "pointrange",
    size = 0.2
  ) +
  theme_minimal() +
  labs(
    x = "Condition",
    y = "Negative affect duration"
  )



p_affectNegative_helping
```




```r
library(lme4)

# affectPositive_categoricalChoice ~ helping*social + (helping | subj_id) + (helping | lab_id)
m_affPos_cat_full <- glmer(
  affectPositive_categoricalChoice ~ helping * social +
    (1 | subj_id),
  data   = trial_level_affect,
  family = binomial
)

print("affectPositive_categoricalChoice ~ helping*social:")
```

```
## [1] "affectPositive_categoricalChoice ~ helping*social:"
```

```r
apa_print(m_affPos_cat_full)$table
```

```
## A data.frame with 5 labelled columns:
##
##                                term estimate       conf.int statistic p.value
## 1                         Intercept    -2.05 [-2.61, -1.48]     -7.11  < .001
## 2                       Helpinghelp    -0.06  [-0.60, 0.48]     -0.21    .836
## 3                      Socialsocial     0.12  [-0.64, 0.88]      0.31    .758
## 4 Helpinghelp $\\times$ Socialsocial     0.34  [-0.42, 1.09]      0.87    .383
##
```

```
## term     : Term
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $z$
## p.value  : $p$
```

```r
# affectNegative_categoricalChoice ~ helping*social + (helping | subj_id) + (helping | lab_id)
m_affNeg_cat_full <- glmer(
  affectNegative_categoricalChoice ~ helping * social +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)

print("affectNegative_categoricalChoice ~ helping*social:")
```

```
## [1] "affectNegative_categoricalChoice ~ helping*social:"
```

```r
apa_print(m_affNeg_cat_full)$table
```

```
## A data.frame with 5 labelled columns:
##
##                                  term estimate       conf.int statistic p.value
## 1                           Intercept    -1.92 [-2.86, -0.98]     -3.99  < .001
## 2                         Helpinghelp    -0.35  [-0.88, 0.18]     -1.31    .190
## 3                        Socialsocial    -0.33  [-1.11, 0.44]     -0.85    .398
## 4 Helpinghelp $\\times$ Socialsocial     0.61  [-0.15, 1.37]      1.57    .117
##
## term     : Term
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $z$
## p.value  : $p$
```

```r
# surprise_categoricalChoice ~ helping*social + (helping | subj_id) + (helping | lab_id)
m_surprise_cat_final <- glmer(
  surprise_categoricalChoice ~ helping*social +
    (1 | subj_id),
  data   = trial_level_affect,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)

print("surprise_categoricalChoice ~ helping*social:")
```

```
## [1] "surprise_categoricalChoice ~ helping*social:"
```

```r
apa_print(m_surprise_cat_final)$table
```

```
## A data.frame with 5 labelled columns:
```

```
##
##                                  term estimate       conf.int statistic p.value
## 1                          Intercept    -2.45 [-3.15, -1.75]     -6.89  < .001
## 2                        Helpinghelp     0.19  [-0.39, 0.77]      0.64    .521
## 3                       Socialsocial     0.18  [-0.72, 1.09]      0.40    .689
## 4 Helpinghelp $\\times$ Socialsocial    -0.28  [-1.10, 0.55]     -0.66    .512
##
## term     : Term
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $z$
## p.value  : $p$
```

```
# chose_helper ~affect_average*helping + (affect_average*helping | participant_id) + (affect_average*he
m_choseHelper_affectAverage_helping <- glmer(
  chose_helper ~ affect_average * helping +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)

print("chose_helper ~affect_average*helping:")
```

```
## [1] "chose_helper ~affect_average*helping:"
```

```
apa_print(m_choseHelper_affectAverage_helping)$table
```

```
## A data.frame with 5 labelled columns:
##
##                                    term estimate      conf.int statistic p.value
## 1                            Intercept    14.81 [9.43, 20.19]      5.40  < .001
## 2                        Affect average    -0.33 [-5.70, 5.04]     -0.12    .905
## 3                           Helpinghelp     0.01 [-2.82, 2.84]      0.01    .993
## 4 Affect average $\\times$ Helpinghelp     0.29 [-6.68, 7.25]      0.08    .935
##
## term     : Term
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $z$
## p.value  : $p$
```

```
# chose_helper ~affectPositive_duration*helping + (affectPositive_duration*helping | participant_id) +
m_choseHelper_affectPositive_helping <- glmer(
  chose_helper ~ affectPositive_duration * helping +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)

print("chose_helper ~affectPositive_duration*helping:")
```

```
## [1] "chose_helper ~affectPositive_duration*helping:"
```

```r
apa_print(m_choseHelper_affectPositive_helping)$table
```

```
## A data.frame with 5 labelled columns:
##
##                                            term estimate        conf.int
## 1                                     Intercept    14.86   [9.54, 20.18]
## 2                       AffectPositive duration    -0.89 [-13.22, 11.44]
## 3                                    Helpinghelp    -0.04   [-2.97, 2.89]
## 4 AffectPositive duration $\\times$ Helpinghelp     0.76 [-14.31, 15.82]
##   statistic p.value
## 1      5.47  < .001
## 2     -0.14    .887
## 3     -0.02    .981
## 4      0.10    .921
##
## term     : Term
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $z$
## p.value  : $p$
```

```r
#chose_helper ~affectNegative_duration*helping + (affectNegative_duration*helping | participant_id) + (
m_choseHelper_affectNegative_helping <- glmer(
  chose_helper ~ affectNegative_duration * helping +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)

print("chose_helper ~affectNegative_duration*helping:")
```

```
## [1] "chose_helper ~affectNegative_duration*helping:"
```

```r
apa_print(m_choseHelper_affectNegative_helping)$table
```

```
## A data.frame with 5 labelled columns:
##
##                                            term estimate        conf.int
## 1                                     Intercept    14.82   [9.41, 20.23]
## 2                       AffectNegative duration     0.01   [-8.12, 8.13]
## 3                                    Helpinghelp     0.02   [-2.99, 3.03]
## 4 AffectNegative duration $\\times$ Helpinghelp    -0.28 [-13.47, 12.92]
##   statistic p.value
## 1      5.37  < .001
## 2      0.00    .998
## 3      0.01    .990
## 4     -0.04    .967
##
## term     : Term
```

```
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $z$
## p.value  : $p$
```

```r
library(lme4)

# chose_helper ~ positive_categoricalChoice*helping + (positive_categoricalChoice*helping | subj_id) +
m_choseHelper_posChoice_full <- glmer(
  chose_helper ~ affectPositive_categoricalChoice * helping +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)

print("chose_helper ~ positive_categoricalChoice*helping:")
```

```
## [1] "chose_helper ~ positive_categoricalChoice*helping:"
```

```r
apa_print(m_choseHelper_posChoice_full)$table
```

```
## A data.frame with 5 labelled columns:
##
##                                                      term estimate      conf.int
## 1                                               Intercept    14.77 [9.43, 20.11]
## 2                       AffectPositive categoricalChoice    -0.06 [-5.53, 5.42]
## 3                                             Helpinghelp    -0.01 [-3.12, 3.09]
## 4 AffectPositive categoricalChoice $\\times$ Helpinghelp    -0.01 [-7.21, 7.18]
##   statistic p.value
## 1      5.42  < .001
## 2     -0.02    .984
## 3     -0.01    .993
## 4      0.00    .997
##
## term     : Term
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $z$
## p.value  : $p$
```

```r
# chose_helper ~ negative_categoricalChoice*helping + (negative_categoricalChoice*helping | subj_id) +
m_choseHelper_negChoice_full <- glmer(
  chose_helper ~ affectNegative_categoricalChoice * helping +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)
print("chose_helper ~ negative_categoricalChoice*helping:")
```

```
## [1] "chose_helper ~ negative_categoricalChoice*helping:"
```

```r
apa_print(m_choseHelper_negChoice_full)$table
```

```
## A data.frame with 5 labelled columns:
##
##                                                      term estimate      conf.int
## 1                                               Intercept    14.76 [9.00, 20.51]
## 2                       AffectNegative categoricalChoice     0.03 [-5.30, 5.37]
## 3                                             Helpinghelp    -0.01 [-3.11, 3.09]
## 4 AffectNegative categoricalChoice $\\times$ Helpinghelp    -0.03 [-7.60, 7.54]
##   statistic p.value
## 1      5.03  < .001
## 2      0.01    .990
## 3     -0.01    .994
## 4     -0.01    .993
##
## term     : Term
## estimate : $\\hat{\\beta}$
## conf.int : 95\\% CI
## statistic: $z$
## p.value  : $p$
```

```r
# chose_helper ~ surprise_categoricalChoice*helping + (surprise_categoricalChoice*helping | subj_id) +
m_choseHelper_surpriseChoice_full <- glmer(
  chose_helper ~ surprise_categoricalChoice * helping +
    (1 | subj_id) +
    (1 | lab_id),
  data   = trial_level_affect,
  family = binomial
)

print("chose_helper ~ surprise_categoricalChoice*helping:")
```

```
## [1] "chose_helper ~ surprise_categoricalChoice*helping:"
```

```r
apa_print(m_choseHelper_surpriseChoice_full)$table
```

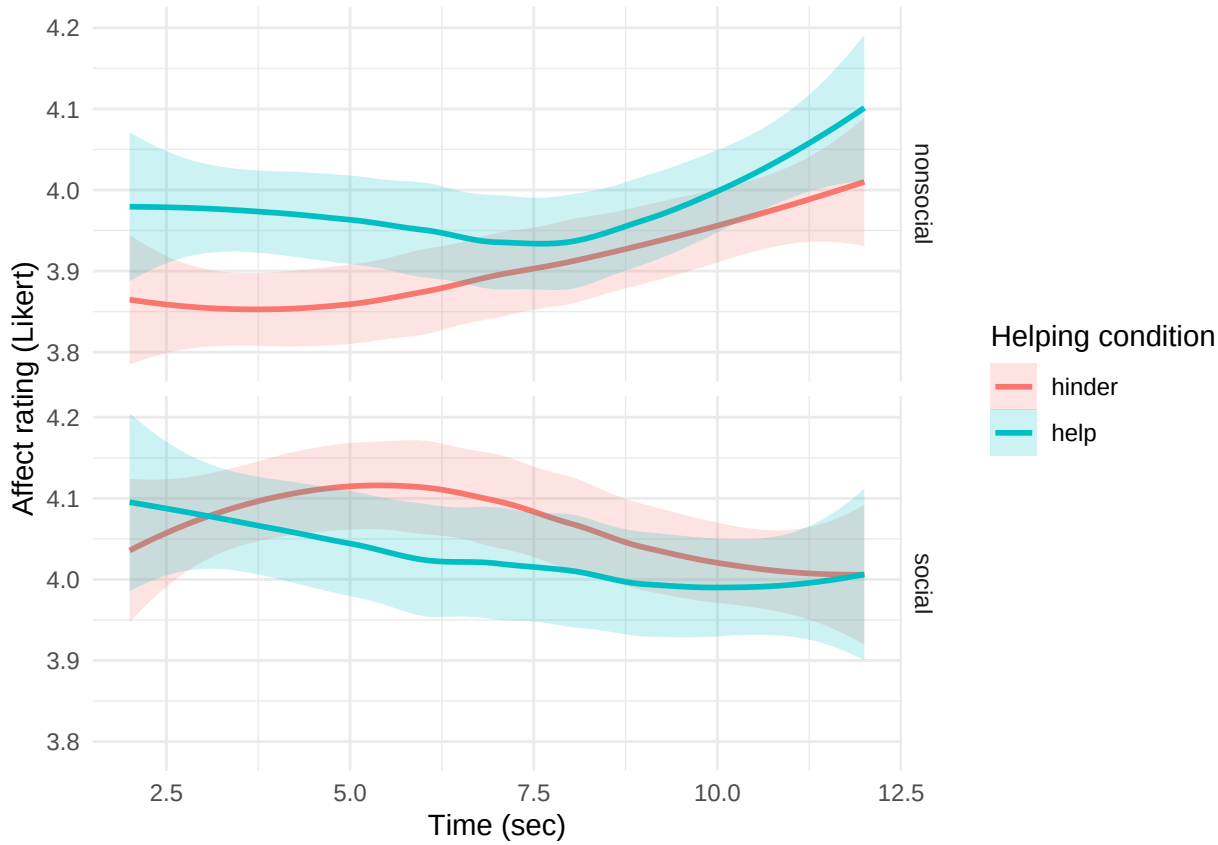
```
## A data.frame with 5 labelled columns:
##
##                                                 term estimate      conf.int
## 1                                          Intercept    14.76 [9.54, 19.99]
## 2                       Surprise categoricalChoice     0.02 [-5.73, 5.77]
## 3                                         Helpinghelp     0.01 [-3.03, 3.05]
## 4 Surprise categoricalChoice $\\times$ Helpinghelp    -0.19 [-8.00, 7.61]
##   statistic p.value
## 1      5.54  < .001
## 2      0.01    .994
## 3      0.01    .994
## 4     -0.05    .961
##
## term     : Term
## estimate : $\\hat{\\beta}$
```

```
## conf.int : 95\\% CI
## statistic: $z$
## p.value  : $p$
```

```r
library(data.table)
setDT(merged_data)

# Exclude rows where start or end time is NA
merged_clean <- merged_data[
  !is.na(segment_start_sec) & !is.na(segment_end_sec)
]

# Expand into 1 Hz time series
time_series_data <- merged_clean[
  ,
  {
    # turn onset and offset into integers
    start_sec <- floor(segment_start_sec)
    end_sec   <- ceiling(segment_end_sec)

    # generate one row per second
    seconds <- seq(start_sec, end_sec, by = 1)

    data.table(
      time_sec = seconds,
      affect_rating_segment = affect_rating_segment,
      lab_id = lab_id,
      subj_id = subj_id,
      folder = folder,
      file_name = file_name,
      segment_start_sec = segment_start_sec,
      segment_end_sec = segment_end_sec,
      duration_sec = duration_sec,
      duration_msec = duration_msec,
      happiness_full_video = happiness_full_video,
      surprise_full_video = surprise_full_video,
      trial_num = trial_num,
      age_days = age_days,
      age_group = age_group,
      condition = condition,
      cb_order = cb_order,
      helper_hinderer_choice = helper_hinderer_choice,
      push_up_identity = push_up_identity,
      push_up_order = push_up_order,
      push_up_side = push_up_side,
      meet_eligibility = meet_eligibility
    )
  },
  by = .I
]


time_series_data[, trial_num_int := as.integer(trial_num)]
time_series_data[
```

```r
  ,
  is_push_up :=
    (push_up_order == "first"  & trial_num_int %% 2L == 1L) |
    (push_up_order == "second" & trial_num_int %% 2L == 0L)
]
time_series_data[
  ,
  helping := factor(
    ifelse(is.na(is_push_up), NA_character_,
           ifelse(is_push_up, "help", "hinder")),
    levels = c("hinder", "help")
  )
]

time_series_data[
  ,
  affect_rating_centered := affect_rating_segment - mean(affect_rating_segment, na.rm = TRUE),
  by = subj_id
]

p_time_affect <- ggplot(
  time_series_data,
  aes(x = time_sec, y = affect_rating_segment, color = helping)
) +
  #geom_jitter(alpha = 0.3, size = 1) +
  geom_smooth(
    method = "loess",
    se = TRUE,
    aes(fill = helping),
    alpha = 0.2
  ) +
  facet_grid(rows = vars(condition)) +
  theme_minimal() +
  labs(
    x = "Time (sec)",
    y = "Affect rating (Likert)",
    color = "Helping condition",
    fill  = "Helping condition"
  )

p_time_affect
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



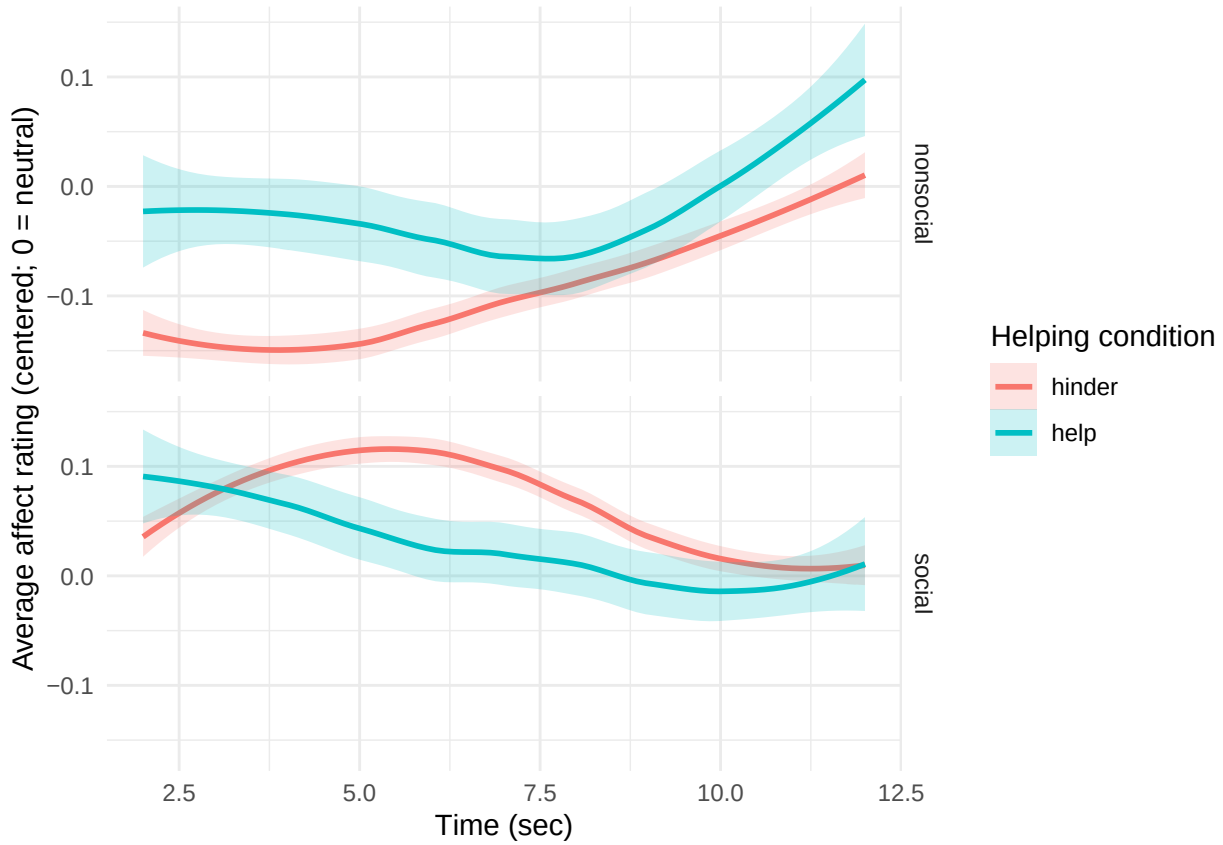

```r
library(data.table)
setDT(merged_data)

# Exclude rows where start or end time is NA
merged_clean <- merged_data[
  !is.na(segment_start_sec) & !is.na(segment_end_sec)
]

# Expand into 1 Hz time series
time_series_data <- merged_clean[
  ,
  {
    # turn onset and offset into integers
    start_sec <- floor(segment_start_sec)
    end_sec   <- ceiling(segment_end_sec)

    # generate one row per second
    seconds <- seq(start_sec, end_sec, by = 1)

    data.table(
      time_sec = seconds,
      affect_rating_segment = affect_rating_segment,

      # Neutral-centered affect (neutral=0). Original scale: 1..7 with 4 = neutral
      affect_c0 = affect_rating_segment - 4,
```

```r
      # Mark each timepoint as 1/0 for proportion measures
      # (mean of these 0/1 variables within a time bin = proportion)
      nonneutral_01 = fifelse(is.na(affect_rating_segment), NA_real_,
                              as.numeric((affect_rating_segment - 4) != 0)),
      pos_01        = fifelse(is.na(affect_rating_segment), NA_real_,
                              as.numeric((affect_rating_segment - 4) > 0)),
      neg_01        = fifelse(is.na(affect_rating_segment), NA_real_,
                              as.numeric((affect_rating_segment - 4) < 0)),

      lab_id = lab_id,
      subj_id = subj_id,
      folder = folder,
      file_name = file_name,
      segment_start_sec = segment_start_sec,
      segment_end_sec = segment_end_sec,
      duration_sec = duration_sec,
      duration_msec = duration_msec,
      happiness_full_video = happiness_full_video,
      surprise_full_video = surprise_full_video,
      trial_num = trial_num,
      age_days = age_days,
      age_group = age_group,
      condition = condition,
      cb_order = cb_order,
      helper_hinderer_choice = helper_hinderer_choice,
      push_up_identity = push_up_identity,
      push_up_order = push_up_order,
      push_up_side = push_up_side,
      meet_eligibility = meet_eligibility
    )
  },
  by = .I
]

time_series_data[, trial_num_int := as.integer(trial_num)]
time_series_data[
  ,
  is_push_up :=
    (push_up_order == "first"  & trial_num_int %% 2L == 1L) |
    (push_up_order == "second" & trial_num_int %% 2L == 0L)
]
time_series_data[
  ,
  helping := factor(
    ifelse(is.na(is_push_up), NA_character_,
           ifelse(is_push_up, "help", "hinder")),
    levels = c("hinder", "help")
  )
]

# Summarize into 1s bins to get time course measures:
# - Avg rating (centered; neutral=0)
# - Proportion not neutral (non-zero)
```

```r
# - Proportion positive / negative
timecourse_1s <- time_series_data[
  ,
  .(
    avg_rating_c0   = mean(affect_c0, na.rm = TRUE),
    prop_nonneutral = mean(nonneutral_01, na.rm = TRUE),
    prop_pos        = mean(pos_01, na.rm = TRUE),
    prop_neg        = mean(neg_01, na.rm = TRUE),
    n_obs           = sum(!is.na(affect_c0))
  ),
  by = .(condition, helping, time_sec)
]

# Plot average affect rating over time (centered; 0 = neutral)
p_time_affect_avg <- ggplot(
  timecourse_1s,
  aes(x = time_sec, y = avg_rating_c0, color = helping)
) +
  #geom_jitter(alpha = 0.3, size = 1) +
  geom_smooth(
    method = "loess",
    se = TRUE,
    aes(fill = helping),
    alpha = 0.2
  ) +
  facet_grid(rows = vars(condition)) +
  theme_minimal() +
  labs(
    x = "Time (sec)",
    y = "Average affect rating (centered; 0 = neutral)",
    color = "Helping condition",
    fill  = "Helping condition"
  )

# Plot proportion non-neutral over time
p_time_affect_prop_nonneutral <- ggplot(
  timecourse_1s,
  aes(x = time_sec, y = prop_nonneutral, color = helping)
) +
  #geom_jitter(alpha = 0.3, size = 1) +
  geom_smooth(
    method = "loess",
    se = TRUE,
    aes(fill = helping),
    alpha = 0.2
  ) +
  facet_grid(rows = vars(condition)) +
  theme_minimal() +
  labs(
    x = "Time (sec)",
    y = "Proportion non-neutral",
    color = "Helping condition",
    fill  = "Helping condition"
```

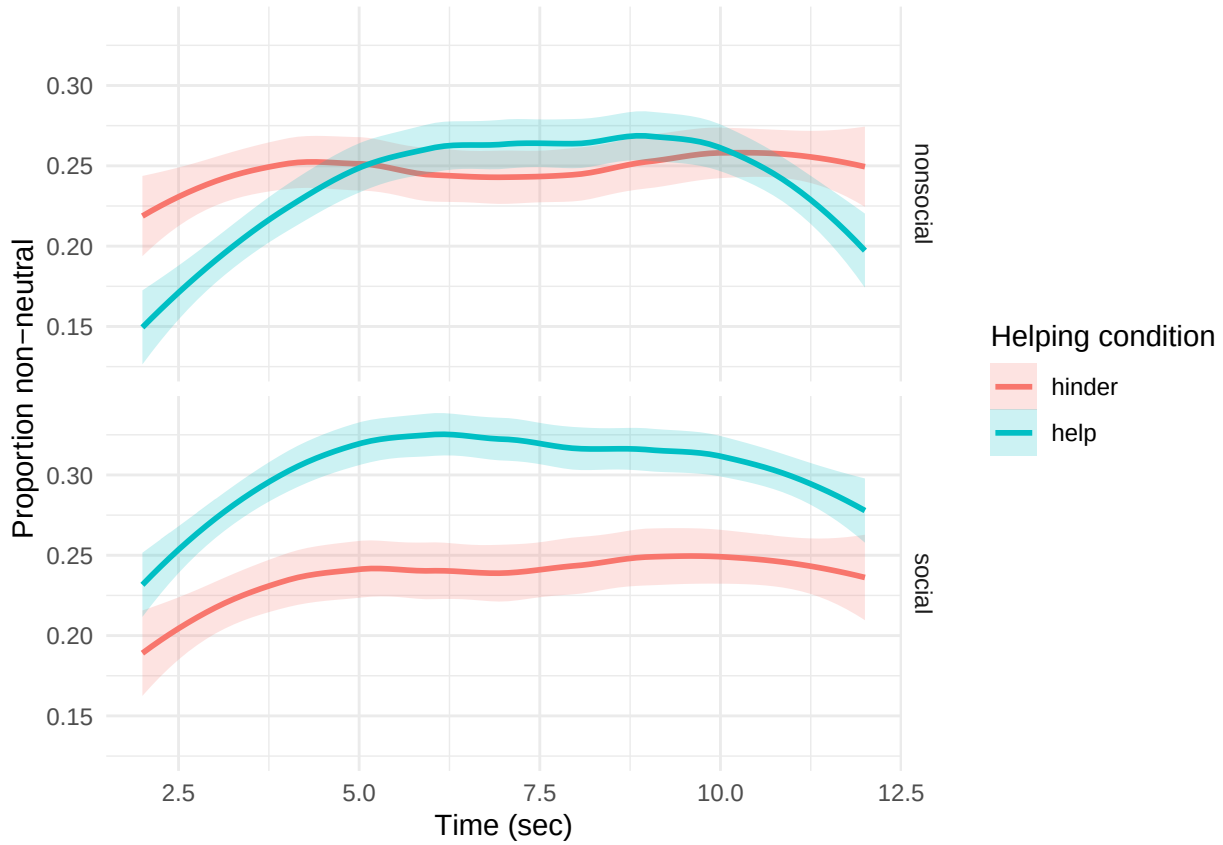
```r
  )

# Plot proportion positive over time
p_time_affect_prop_pos <- ggplot(
  timecourse_1s,
  aes(x = time_sec, y = prop_pos, color = helping)
) +
  #geom_jitter(alpha = 0.3, size = 1) +
  geom_smooth(
    method = "loess",
    se = TRUE,
    aes(fill = helping),
    alpha = 0.2
  ) +
  facet_grid(rows = vars(condition)) +
  theme_minimal() +
  labs(
    x = "Time (sec)",
    y = "Proportion positive",
    color = "Helping condition",
    fill  = "Helping condition"
  )

# Plot proportion negative over time
p_time_affect_prop_neg <- ggplot(
  timecourse_1s,
  aes(x = time_sec, y = prop_neg, color = helping)
) +
  #geom_jitter(alpha = 0.3, size = 1) +
  geom_smooth(
    method = "loess",
    se = TRUE,
    aes(fill = helping),
    alpha = 0.2
  ) +
  facet_grid(rows = vars(condition)) +
  theme_minimal() +
  labs(
    x = "Time (sec)",
    y = "Proportion negative",
    color = "Helping condition",
    fill  = "Helping condition"
  )

# Print plots
p_time_affect_avg
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



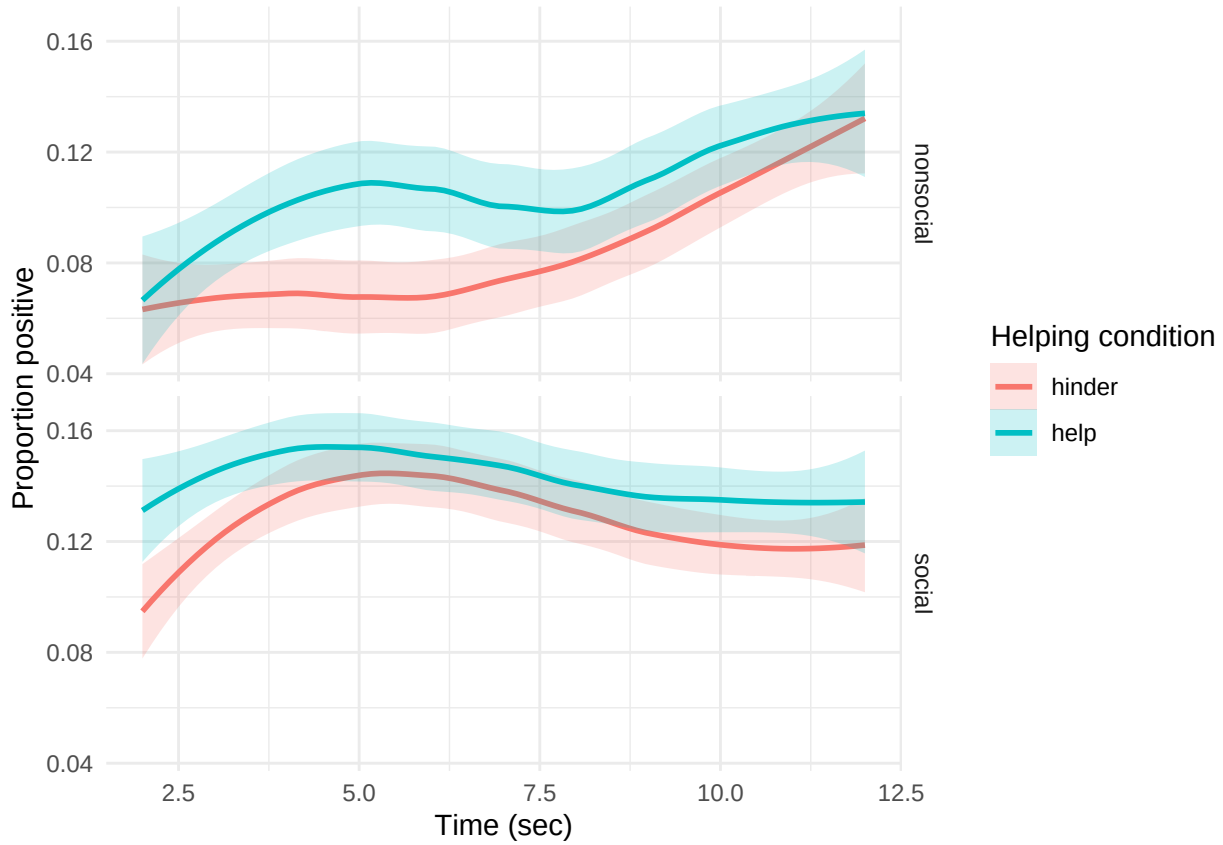

```
p_time_affect_prop_nonneutral
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



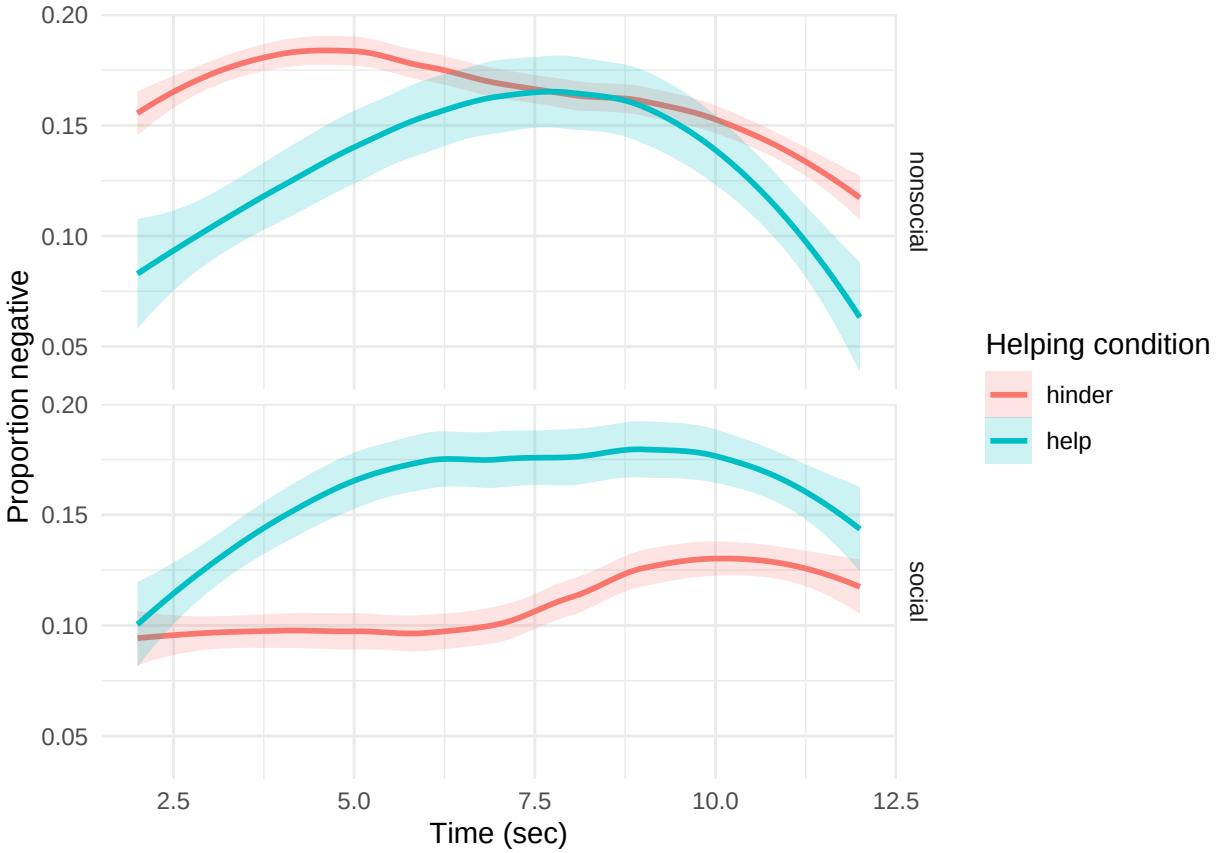

```
p_time_affect_prop_pos
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
p_time_affect_prop_neg
```

```
## `geom_smooth()` using formula = 'y ~ x'
```


0.20
0.15
0.10
0.05
Proportion negative
nonsocial
social
2.5
5.0
7.5
10.0
12.5
Time (sec)
Helping condition
hinder
help